Clustering, Prototype Narratives, and Stiefel-Manifold Orthogonalization:

A Worked Methodology on CNN Live-Blog Coverage of the 2026 Iran War



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Following the Anglade-Lehalle-Massi approach of orthogonalisation of narratives
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Positioning: What Are We Talking About?

The problem

A corpus of **1,572** paragraph-level snippets from CNN's live-blog coverage of the US–Israel–Iran conflict (March–August 2026). We want to:

- discover the **narrative themes** present in the corpus,
- build a **clean, discriminative vector** representing each theme,
- and track any document's **exposure** to a theme over time – without that exposure being contaminated by generic "war-reporting English".

The idea, in one sentence

Cluster in the natural embedding space, build a short hand-written **prototype** per cluster, then **orthogonalize** the prototypes on the Stiefel manifold to strip out what all themes share, leaving a **pure narrative** per theme.



Time to Conclude, Read Ahead: The Take-Away

Overview of the Pipeline

Stage 1: Embedding & Clustering

Stage 2: Representative Narratives & Prototypes

Stage 3: Stiefel-Manifold Orthogonalization

Stage 4: Scoring Under Three Representations

Stage 5: Ex Post Validation



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Five Stages, From Raw Text to Validated Pure Narratives

1. **Embed & cluster** the corpus in its natural, untouched embedding space ($K = 8$ clusters, K-Means on PCA scores).
2. **Curate & average** three representative narratives per cluster into one **prototype of narrative**.
3. **Orthogonalize** the eight prototypes on the Stiefel manifold $\rightarrow$ **pure narratives**, purged of shared background noise.
4. **Score** every snippet against three representations: original / pure (orthogonal) / residual.
5. **Validate ex post**: does the pure representation separate in-cluster from out-of-cluster documents *better* than the raw one?

Why this matters
If pure narratives cleanly separate documents, then tracking a document's cosine similarity to a pure narrative *over time* reflects genuine shifts in **exposure to that narrative** – not drift in how much generic conflict vocabulary a day's reporting happens to use.

Stage 1: Embedding & Clustering

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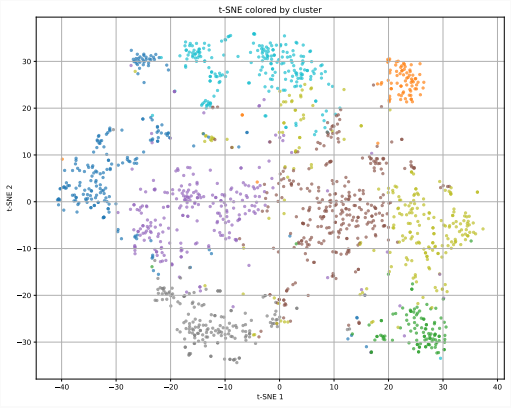
Stage 5: Ex Post Validation



Embedding and Clustering in the Natural Space

- $n = 1{,}572$ snippets, March 1 – August 18, 2026
- Each snippet $\rightarrow$ unit-norm $\mathbf{e}_i \in \mathbb{R}^{768}$ (`nomic-ai/modernbert-embed-base`)
- $\mathrm{sim}(s_i, s_j) = \mathbf{e}_i^\top \mathbf{e}_j$
- PCA to 50 dims (numerical stability only)
- K-Means, $K = 8$, fit on the PCA scores

Key methodological point
No orthogonalization or transformation is applied before or during clustering: whatever structure K-Means finds is a property of the embedding model and the corpus alone, not an artifact of the machinery introduced later.



t-SNE projection (diagnostic only), colored by K-Means cluster.

The Eight Clusters

#	Theme	Size
0	Gulf Regional Military Escalation	209
1	Global Energy Markets & Economic Fallout	81
2	Iranian Political Transition & Internal Affairs	110
3	US-Israel Military Campaign in Iran	246
4	US Domestic Politics & Presidential Leadership	306
5	Israel-Hezbollah-Lebanon Theater	170
6	Diplomacy & International Mediation	235
7	**Maritime Conflict & Strait of Hormuz**	215
	Total	**1,572**

Worked example used throughout: **Maritime Conflict & Strait of Hormuz** – shipping disruption, tanker attacks, naval blockades, transit through the Strait.



Stage 2: Representative Narratives & Prototypes

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From Representative Narratives to a Prototype

Worked example – Maritime Conflict & Strait of Hormuz

- $r_{7,1}$: Maritime traffic through the Strait of Hormuz has slowed to a near halt as shipping companies reroute or suspend passage.
- $r_{7,2}$: US forces intercepted and seized a cargo vessel attempting to bypass the naval blockade of Iranian ports.
- $r_{7,3}$: An oil tanker was struck and disabled by an unidentified projectile near the strait's entrance.

Three hand-picked, illustrative sentences per cluster (curated by the analyst, *not* the medoid/centroid), embedded and averaged:

$$N_i = \frac{\frac{1}{3}\sum_{k=1}^{3} r_{i,k}}{\left\| \frac{1}{3}\sum_{k=1}^{3} r_{i,k} \right\|}, \qquad i = 1, \ldots, 8$$

- Encodes the analyst's semantic understanding of the theme, independently of how K-Means partitioned the data.
- Averaging cancels idiosyncratic wording, keeps the common direction.



Stage 3: Stiefel-Manifold Orthogonalization

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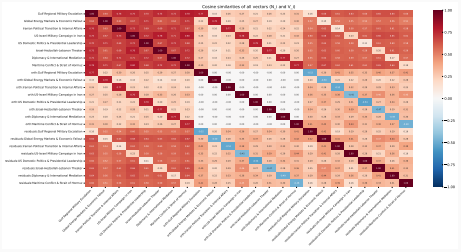
Stage 4: Scoring Under Three Representations

Stage 5: Ex Post Validation



The Problem: Prototypes Are Highly Correlated

- Pairwise cosine similarity between the 8 prototypes: typically **0.6 – 0.8** off-diagonal.
- Interpretation: a shared **background noise** –
  - ordinary English grammar,
  - the generic vocabulary of Iran-war reporting (place names, military/diplomatic jargon, "Iran", "strikes"...).
- Consequence: a snippet can score highly against several prototypes simply because the prototypes themselves overlap – not because the snippet is thematically ambiguous.



Cosine similarity: originals (top-left), orthogonal (middle), residuals (bottom-right).

Goal
Strip the shared component out, isolate the **pure narrative** of each theme.

The Optimization Problem on the Stiefel Manifold

Find $V \in \mathrm{St}(K, d) = \{V \in \mathbb{R}^{K \times d} : VV^\top = I_K\}$, as close as possible to N, by solving a worst-case alignment problem:

$$\max_{V \in \mathrm{St}(K,d)} \min_{i=1,\dots,K} \langle N_i, V_i \rangle$$

- No single narrative is sacrificed to improve the others.
- Smoothed via a soft-min surrogate (temperature $\tau = 0.05$):

$$\mathrm{softmin}_\tau(x_1, \dots, x_K) = -\tau \log \sum_{i=1}^{K} \exp\left(-\frac{x_i}{\tau}\right)$$

- Minimized with **Riemannian conjugate gradient** (pymanopt), initialized by Gram–Schmidt on N.
- Fast: 24–40 iterations, a few seconds.

Three comparable objects per cluster i

Prototype N_i (signal + shared noise) $\rightarrow$ **Pure narrative** V_i (mutually orthogonal by construction) $\rightarrow$ **Residual** $R_i = \dfrac{N_i - V_i}{\|N_i - V_i\|}$ (the removed background noise)

Worked Example: From Prototype to Pure Narrative (Maritime)

- N_{Maritime} has cosine similarity ≈ 0.78 with $N_{\text{US-Israel Campaign}}$ and ≈ 0.71 with $N_{\text{Diplomacy}}$ – shared "military operations" / "negotiation" vocabulary.
- After optimization, V_{Maritime} has, **by construction**, cosine similarity ≈ 0.00 with every other pure narrative, while retaining alignment ≈ 0.77 with its own original direction N_{Maritime}: still recognizably "maritime conflict", just stripped of what it shared with the rest.
- The residual R_{Maritime} instead correlates **positively** with the other clusters' prototypes (0.51–0.67) and **negatively** with its own pure narrative (-0.48): exactly the shared, non-discriminative component removed by orthogonalization.

Stage 4: Scoring Under Three Representations

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Three Similarity Matrices per Snippet

Every snippet s_t carries a K-Means label $c(t)$ from Stage 1 (obtained *before* orthogonalization existed). It is scored against all 8 prototypes under each representation:

$$S^{\text{orig}}_{t,i} = \mathbf{e}_t^\top N_i, \qquad S^{\text{orth}}_{t,i} = \mathbf{e}_t^\top V_i, \qquad S^{\text{res}}_{t,i} = \mathbf{e}_t^\top R_i$$

Worked example – one Maritime snippet

Prototype (j)	S^{orig}	S^{orth}	S^{res}
Maritime Conflict (own cluster)	~ 0.80	~ 0.52	~ 0.30
US-Israel Military Campaign	~ 0.75	~ 0.28	~ 0.55
Diplomacy & Mediation	~ 0.70	~ 0.22	~ 0.50

Under original: own theme wins by a narrow margin (crowded space). **Under orthogonal:** the margin widens sharply once shared signal is projected out – unambiguously closest to its purified theme.

Stage 5: Ex Post Validation

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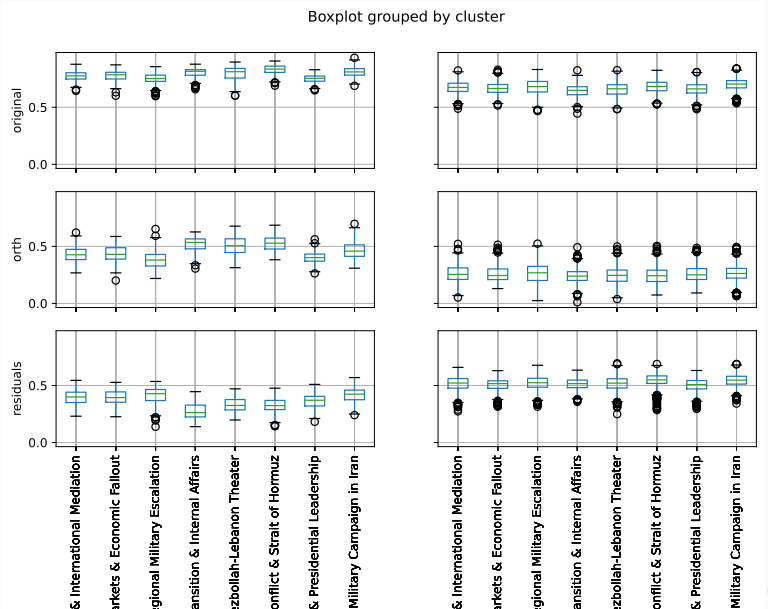
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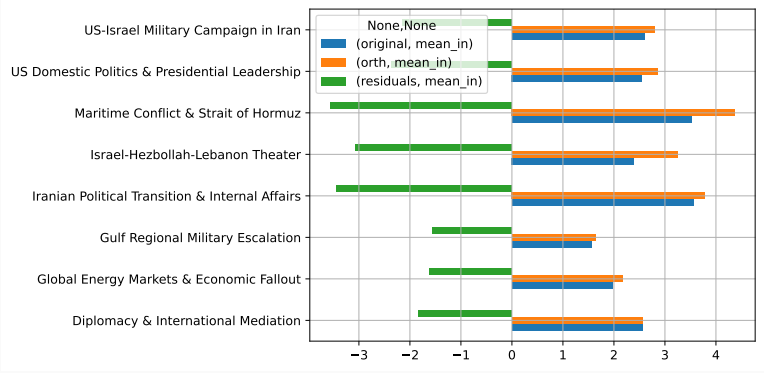
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Does Orthogonalization Improve Discrimination?



Standardized Separation, Theme by Theme



$(\bar{s}_{\text{in}} - \bar{s}_{\text{out}})/\sigma_{\text{in}}$ for each theme. Orange (orth) > blue (original) for every single theme; green (residuals) is uniformly negative.

Maritime Conflict: largest gain of all themes – $\approx 4.3\,\sigma$ (orth) vs. $\approx 3.5\,\sigma$ (original) – a technical vocabulary ("Strait of Hormuz", "tanker", "blockade") survives orthogonalization essentially intact.

Take-Away

Across all eight themes, the **pure-narrative** representation **matches or exceeds** the raw prototype's discriminative power, while the **residual** representation is **uniformly non-discriminative**.

Since pure narratives come from only 3 hand-picked sentences per cluster and an unsupervised geometric constraint (orthonormality) – *not* from the K-Means labels – and since K-Means itself ran on the untouched embedding space, this is an **unsupervised, label-free corroboration** that:

1. the 8 K-Means clusters correspond to **genuinely separable** narrative axes;
2. Stiefel orthogonalization recovers **additional, meaningful discriminative signal** masked by shared background noise;
3. tracking a document's (or a day's) similarity to a **pure narrative over time** is a meaningful way to monitor exposure to that narrative, largely free of confounding drift in generic war-reporting language.



Thank you for your attention



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Appendix: The Eight Clusters, in Detail i

Cluster 0 – Gulf Regional Military Escalation ($n = 209$) Direct attacks, drone/missile interceptions, evacuations across secondary Gulf states (Saudi Arabia, Kuwait, UAE, Bahrain, Qatar, Iraq, Yemen).

Cluster 1 – Global Energy Markets & Economic Fallout ($n = 81$) Oil-price surges, market volatility, SPR releases and other economic consequences of the war.

Cluster 2 – Iranian Political Transition & Internal Affairs ($n = 110$) Internal governance, death of Supreme Leader Khamenei, succession, domestic repression.

Cluster 3 – US-Israel Military Campaign in Iran ($n = 246$) Direct kinetic operations – airstrikes, missile exchanges, naval engagements, CENTCOM updates.

Cluster 4 – US Domestic Politics & Presidential Leadership ($n = 306$) White House rhetoric, press briefings, congressional oversight debates.

Cluster 5 – Israel-Hezbollah-Lebanon Theater ($n = 170$) Parallel Israel–Hezbollah/Lebanon front: rocket/airstrike exchanges, fragile ceasefires.

Cluster 6 – Diplomacy & International Mediation ($n = 235$) Diplomatic backchannels (Qatar, Pakistan, Oman), ceasefire/peace frameworks.

Cluster 7 – Maritime Conflict & Strait of Hormuz ($n = 215$) Shipping disruption, naval blockades, tanker attacks, transit through the Strait of Hormuz.

